

Spectral Bounds for Classical & Quantum Graph Parameters

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Spectral graph theory

- ▶ Spectral graph theory is a well-established branch of graph theory.
- ▶ It studies properties of graphs relying on eigenvalues and eigenvectors of matrices associated to graphs such as their adjacency and Laplacian matrices.
- ▶ One of the most well-known results in graph theory is the Hoffman lower bound on the chromatic number.

Graph parameters I

Chromatic numbers and Lovász theta functions

- ▶ $\omega(G)$ – clique number
- ▶ $\chi_v(G)$ – vector chromatic number
- ▶ $\chi_{sv}(G)$ – strict vector chromatic number
- ▶ $\chi_{rv}(G)$ – rigid vector chromatic number
- ▶ $\chi_{\text{vect}}(G)$ – vectorial chromatic number
- ▶ $\theta'(\overline{G})$ – Schrijver variant of Lovász theta function
- ▶ $\theta(\overline{G})$ – Lovász theta function
- ▶ $\theta^+(\overline{G})$ – Szegedy variant of Lovász theta function
- ▶ These parameters, other than $\omega(G)$ can all be calculated using semi-definite programming.

Graph parameters II

- ▶ $\xi_f(G)$ – projective rank
- ▶ $\xi(G)$ – orthogonal rank
- ▶ $\xi'(G)$ – normalized orthogonal rank
- ▶ $\chi_q(G)$ – quantum chromatic number
- ▶ $\chi_q^{(1)}(G)$ – rank one quantum chromatic number
- ▶ $\chi_f(G)$ – fractional chromatic number
- ▶ $\chi_c(G)$ – circular chromatic number
- ▶ $\chi(G)$ – chromatic number

Hierarchies of graph parameters

$$\omega(G) \longrightarrow \chi_v(G) = \theta'(\overline{G}) \longrightarrow \chi_{sv}(G) = \theta(\overline{G}) \longrightarrow \chi_{rv}(G) = \theta^+(\overline{G}) \longrightarrow \xi_f(G)$$

$\lceil \theta^+(\overline{G}) \rceil = \chi_{\text{vect}}(G)$

$$\begin{array}{ccccccc} \chi_{\text{vect}}(G) & \longrightarrow & \xi(G) & & & & \\ & \searrow & \nearrow & \searrow & & & \\ \xi_f(G) & \longrightarrow & \chi_q(G) & \longrightarrow & \chi_q^{(1)}(G) & \longrightarrow & \xi'(G) \longrightarrow \lceil \chi_c(G) \rceil = \chi(G) \\ & \searrow & & & & & \\ & & \chi_f(G) & \longrightarrow & \chi_c(G) & \nearrow & \end{array}$$

Here $\longrightarrow$ is used to indicate the less or equal relationship.

Coloring & chromatic number

- ▶ We consider only undirected graphs without multiple edges and without loops.
- ▶ A **c-coloring** of a graph $G = (V, E)$ is a collection of colors

$$\{c_v : v \in V\}$$

in $[c] = \{0, \dots, c-1\}$ such that for all $vw \in E$

$$c_v \neq c_w.$$

- ▶ The **chromatic number** χ of a graph G is the minimum c admitting a c -coloring of G .
- ▶ The chromatic number is NP-hard to approximate.

Quantum coloring, quantum chromatic number

- ▶ A **quantum c -coloring** of a graph $G = (V, E)$ is a collection of orthogonal projectors

$$\{P_{v,k} : v \in V, k \in [c]\}$$

acting on $\mathbb{C}^d$ for some $d > 0$ such that

- ▶ for all vertices $v \in V$

$$\sum_{k \in [c]} P_{v,k} = I_d \quad \text{completeness}$$

- ▶ for all edges $vw \in E$ and for all $k \in [c]$

$$P_{v,k} P_{w,k} = 0_d \quad \text{orthogonality}$$

- ▶ The **quantum chromatic number** χ_q is the smallest c for which the graph G admits a quantum c -coloring.

Classical coloring as a special case of quantum coloring

- ▶ Let $\{c_v : v \in V\}$ be a collection of colors in $[c]$ that defines a classical c -coloring.
- ▶ For $v \in V$, consider the “one-hot-encoding”

$$\{P_{v,k} : k \in [c]\}$$

of the color $c_v \in [c]$, that is,

$$P_{v,k} = [c_v = k],$$

where $[\cdot]$ denotes the Iverson bracket.

- ▶ For instance, if $c = 3$ and some vertex v is colored with color $c_v = 2$, then the corresponding one-hot-encoding is given by

$$P_{v,0} = 0, \quad P_{v,1} = 0, \quad P_{v,2} = 1.$$

Classical coloring as a special case of quantum coloring

- ▶ It is immediate that a classical c -coloring gives rise to a quantum c -coloring coloring in dimension $d = 1$.
- ▶ The orthogonal projectors in the resulting quantum c -coloring are the scalars 0 and 1.

Exponential separation between classical and quantum chromatic numbers

- ▶ The **orthogonality graph**, $\Omega(n)$, has the vertex set of ± 1 -vectors of length n , with two vertices adjacent if they are orthogonal.
- ▶ The **quantum** chromatic number is **linear** in n (equal to n), whereas the **classical** chromatic number $\chi(\Omega(n))$ is **exponential** in n .

Hoffman lower bound on the quantum chromatic number

- ▶ Let $A = (a_{vw}) \in \mathbb{C}^{n \times n}$ denote the adjacency matrix of a graph G , that is,
 - ▶ $a_{vw} = 1$ for all $vw \in E$
 - ▶ $a_{vw} = 0$ otherwise
- ▶ Let $\mu_{\max}$ and $\mu_{\min}$ denote the maximum and minimum eigenvalues of A , respectively.
- ▶ The **Hoffman bound** states

$$\chi_q \geq 1 + \frac{\mu_{\max}}{|\mu_{\min}|},$$

Quantum coloring $\Rightarrow$ matrix equality for the adjacency matrix

- We will show later that starting from a quantum c -coloring we can construct a unitary matrix

$$U \in \mathbb{C}^{n \times n} \otimes \mathbb{C}^{d \times d}$$

such that the following **matrix equality** holds

$$\sum_{\ell=1}^{c-1} U^\ell (-A \otimes I_d) (U^\dagger)^\ell = A \otimes I_d.$$

- Once we have established the above matrix equality for the adjacency matrix, we can use results from **majorization** theory to prove the Hoffman bound.

Majorization inequality for maximum eigenvalues

- ▶ Let X and Y be two arbitrary hermitian matrices in $\mathbb{C}^{m \times m}$.
- ▶ It is well known that the maximum eigenvalues of the matrices satisfy

$$\mu_{\max}(X) + \mu_{\max}(Y) \geq \mu_{\max}(X + Y).$$

Proof of the Hoffman bound

- ▶ Applying (i) the majorization bound and (ii) the invariance of the spectrum under conjugation with unitaries to the matrix equality for the adjacency matrix

$$\sum_{\ell=1}^{c-1} U^\ell (-A \otimes I_d) (U^\dagger)^\ell = A \otimes I_d$$

yields

$$(c-1)\mu_{\max}(-A \otimes I_d) \geq \mu_{\max}(A \otimes I_d)$$

- ▶ This further simplifies to

$$(c-1)|\mu_{\min}(A)| \geq \mu_{\max}(A)$$

which is the Hoffman lower bound.

Inertial lower bound on the quantum chromatic number

- ▶ Let (n^+, n^0, n^-) denote the **inertia** of a graph G , that is, the number of positive, zero, and negative eigenvalues of its adjacency matrix A .
- ▶ The **inertial bound** states

$$\chi_q \geq 1 + \max \left\{ \frac{n^+}{n^-}, \frac{n^-}{n^+} \right\}.$$

Rank inequality for positive semidefinite matrices

- ▶ A hermitian matrix $X \in \mathbb{C}^{m \times m}$ is called **positive semidefinite** if all its eigenvalues are nonnegative.
- ▶ Let $X, Y \in \mathbb{C}^{m \times m}$ be two hermitian matrices.

We say that X is **greater or equal to** Y , denoted by $X \geq Y$, if their difference $X - Y$ is positive semidefinite.

- ▶ Assume that $X, Y \in \mathbb{C}^{m \times m}$ are two positive semidefinite matrices with $X \geq Y$.
- ▶ Then, we have the rank inequality

$$\text{rank}(X) \geq \text{rank}(Y).$$

Decomposition of the adjacency matrix in positive and negative parts

- ▶ Let $\mu_1 \geq \mu_2 \geq \dots \geq \mu_n$ denote the eigenvalues of the adjacency matrix A and $P_1, \dots, P_n$ the corresponding one-dimensional projectors.
- ▶ Define

$$A^+ = \sum_{i:\lambda_i > 0} \mu_i P_i \qquad P^+ = \sum_{i:\lambda_i > 0} P_i$$

$$A^- = \sum_{i:\lambda_i < 0} |\mu_i| P_i \qquad P^- = \sum_{i:\lambda_i < 0} P_i$$

- ▶ Then, we can write

$$A = A^+ - A^-$$

- ▶ Note that

$$n^\pm = \text{rank}(A^\pm) \quad \text{and} \quad A^\pm = \pm P^\pm A P^\pm$$

$$\text{and also } n = n^+ + n^0 + n^-.$$

Proof of the inertial bound

- ▶ We start with the matrix equality

$$\sum_{\ell=1}^{c-1} U^\ell (-A \otimes I_d) (U^\dagger)^\ell = A \otimes I_d$$

- ▶ Plugging in $A = A^+ - A^-$ yields

$$\sum_{\ell=1}^{c-1} U^\ell ((A^- - A^+) \otimes I_d) (U^\dagger)^\ell = (A^+ - A^-) \otimes I_d$$

- ▶ Multiplying both sides by P^+ from left and right yields

$$\sum_{\ell=1}^{c-1} P^+ U^\ell ((A^- - A^+) \otimes I_d) (U^\dagger)^\ell P^+ = A^+ \otimes I_d$$

Proof of the inertial bound

- ▶ Dropping the subtraction on the left hand side yields

$$\sum_{\ell=1}^{c-1} P^+ U^\ell (A^- \otimes I_d) (U^\dagger)^\ell P^+ \geq A^+ \otimes I_d$$

- ▶ Using the rank inequality and some elementary results yields

$$(c-1) \cdot \text{rank}(A^- \otimes I_d) \geq \text{rank}(A^+ \otimes I_d)$$

which simplifies to the inertial bound

$$c \geq 1 + \frac{n^+}{n^-}.$$

The other inequality $c \geq 1 + n^-/n^+$ is proved similarly.

Proof of matrix inequality

- In the following, we will prove the matrix equality

$$\sum_{\ell=1}^{c-1} U^{\ell}(-A \otimes I_d)(U^{\dagger})^{\ell} = A \otimes I_d$$

with the help of **pinching** and **twirling**.

Pinching

- ▶ Let $\{Q_k : k \in [c]\}$ be any collection of orthogonal projectors in $\mathbb{C}^{m \times m}$ that form a resolution of the identity.
- ▶ Then, the operation $\mathcal{C}$ that maps an arbitrary matrix $X \in \mathbb{C}^{m \times m}$ to

$$\mathcal{C}(X) = \sum_{k \in [c]} Q_k X Q_k$$

is called **pinching**.

- ▶ We say that the pinching $\mathcal{C}$ **annihilates** X if

$$\mathcal{C}(X) = 0_m.$$

Pinching from quantum coloring

- Assume that the orthogonal projectors

$$P_{v,k} \quad v \in V, k \in [c]$$

acting on $\mathbb{C}^d$ form a quantum c -coloring of G .

- Then, the block-diagonal orthogonal projectors that act on $\mathbb{C}^n \otimes \mathbb{C}^d$ and are defined by

$$\begin{aligned} P_k &= \bigoplus_{v \in V} P_{v,k} \\ &= \sum_{v \in V} |v\rangle\langle v| \otimes P_{v,k} \quad k \in [c] \end{aligned}$$

give rise to a pinching $\mathcal{C}$.

Pinching from quantum coloring

- ▶ Due to the block diagonal structure of P_k we have

$$P_k(A \otimes I_d)P_k = \sum_{v,w \in V} a_{vw} |v\rangle\langle w| \otimes P_{v,k}P_{w,k}.$$

- ▶ Due to the orthogonality condition we have

$$a_{vw} = 1 \quad \Rightarrow \quad P_{v,k}P_{w,k} = 0_d.$$

- ▶ Therefore, this pinching annihilates $A \otimes I_d$, that is,

$$\mathcal{C}(A \otimes I_d) = 0_n \otimes 0_d.$$

Twirling

- ▶ Let $\{U_\ell : \ell \in [c]\}$ be any collection of unitary matrices in $\mathbb{C}^{m \times m}$.
- ▶ Then, the operation $\mathcal{D}$ that maps an arbitrary matrix $X \in \mathbb{C}^{m \times m}$ to

$$\mathcal{D}(X) = \frac{1}{c} \sum_{\ell \in [c]} U_\ell X U_\ell^\dagger$$

is called **twirling**.

- ▶ We say that the twirling $\mathcal{D}$ **annihilates** X if

$$\mathcal{D}(X) = 0_m.$$

Twirling from pinching

- ▶ Let $\{Q_k : k \in [c]\}$ be a collection of orthogonal projectors defining a pinching operation $\mathcal{C}$.
- ▶ Let $\omega = e^{2\pi i/c}$ be a c th root of unity.
- ▶ Then, the collection of unitary matrices $\{U^\ell : \ell \in [c]\}$ consisting of the powers of the unitary U

$$U = \sum_{k \in [c]} \omega^k Q_k,$$

defines a twirling operation $\mathcal{D}$ such that

$$\mathcal{C}(X) = \mathcal{D}(X)$$

for all matrices $X \in \mathbb{C}^{m \times m}$.

Twirling from quantum coloring

- Use the orthogonal projectors $P_{k,v}$, $v \in V, k \in [c]$, of a quantum c -coloring to construct the corresponding block-diagonal orthogonal projectors

$$P_k = \sum_{v \in V} |v\rangle\langle v| \otimes P_{v,k} \quad k \in [c]$$

acting on $\mathbb{C}^n \otimes \mathbb{C}^d$ and the unitary

$$U = \sum_{k \in [c]} \omega^k P_k.$$

- Then, the twirling map $\mathcal{D}$ defined by the powers of U

$$\mathcal{D}(X) = \frac{1}{c} \sum_{\ell \in [c]} U^\ell X (U^\dagger)^\ell$$

annihilates $A \otimes I_d$. This is equivalent to the matrix equality.

Importance of twirling from quantum coloring

- ▶ The construction of twirling & pinching from quantum coloring leads to a unified way of proving the Hoffman and inertial lower bounds for χ_q .
- ▶ Moreover, any existing or future lower bound on the minimum number of operations for pinching and twirling to annihilate A is automatically a lower bound for χ_q .

Implication of the lower bounds

- ▶ The spectrum of the Clebsch graph on 16 vertices is $(5^1, 1^{10}, -3^5)$.
- ▶ It is known that $\chi = 4$.
- ▶ The Hoffman bound implies $\chi_q \geq 3$, whereas the inertial bounds $\chi_q \geq 4$.
- ▶ Sometimes these and other spectral lower bounds allow us to determine the quantum chromatic number.
- ▶ It is not known whether the quantum chromatic number is computable which is why spectral bounds are particularly useful and interesting.

Simulating Hamiltonians

- ▶ Consider a quantum system $(\mathbb{C}^2)^{\otimes n}$ system with Hamiltonian

$$H = \sum_{k < \ell} \sum_{a,b} c_{ab}^{k\ell} \sigma_a^k \sigma_b^\ell$$

- ▶ Assume that the control operations $U \in \mathcal{U}(2)^{\otimes n}$ are available.
- ▶ Consider of the problem of simulating the Hamiltonian $-H$ by interspersing the natural time with control operations using average Hamiltonian theory.
- ▶ The time complexity of the required control scheme depends on the chromatic number of the graph describing the coupling topology of the system Hamiltonian.
- ▶ I studied Hamiltonian simulation in my PhD and this is how I originally found my way to spectral graph theory.

Joint work with Clive Elphick

-  P. Wocjan and C. Elphick, *Spectral lower bounds for the orthogonal and projective ranks of a graph*, arXiv:1806.02734, Electronic Journal of Combinatorics.
-  P. Wocjan and C. Elphick, *Spectral lower bounds for the quantum chromatic number of a graph*, arXiv:1805.08334, Journal of Combinatorial Theory, Series A.
-  P. Wocjan and C. Elphick, *More Tales of Hoffman: bounds for the vector chromatic number of a graph*, arXiv:1812.02613.
-  P. Wocjan and C. Elphick, *An inertial upper bound for the quantum independence number of a graph*, arXiv:1808.10820.